

Quantifier-Spreading in Korean Children and Adults: Evidence from an Eye-Tracking Study*

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This study examined how Korean-speaking children and adults interpret universally quantified sentences and explored the mechanisms behind quantifier-spreading (Q-spreading) errors. Using an eye-tracking sentence–picture verification task, the research involved 57 preschoolers (aged 5–6) and 97 adults across various visual contexts. The accuracy results indicated that children frequently made Q-spreading errors, particularly in visually complex scenes, while adults performed almost perfectly. Eye-movement data showed that children’s errors were associated with increased fixations on visually salient but truth-conditionally irrelevant objects. This suggests that Q-spreading is driven by attentional competition and limited inhibitory control, rather than a lack of semantic knowledge. These findings support processing-based theories that highlight the importance of executive control and pragmatic reasoning in resolving referential ambiguity. Additionally, the results imply that acquiring universal quantification involves not only semantic comprehension but also the gradual integration of linguistic, cognitive, and pragmatic processes. By demonstrating that gaze allocation serves as a real-time indicator of quantifier integration, this study underscores the methodological significance of eye-tracking in investigating the relationship between language and cognition. Overall, Q-spreading errors represent an intermediate stage in the development of linguistic and cognitive integration.

Keywords Quantifier spreading, eye-tracking, executive control, child language

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I. INTRODUCTION

A substantial body of research has shown that young children frequently make errors when interpreting sentences containing universal quantifiers (e.g., *every* in English). A widely studied error pattern, referred to as quantifier-spreading (Q-spreading), occurs when children incorrectly reject sentences containing *every* in contexts that include additional, unmentioned objects (Brooks & Braine, 1996; Crain et al., 1996; Freeman, 1985; Inhelder & Piaget, 1964; Minai et al., 2012; Philip, 1995; Rakhlin, 2007; Roeper et al., 2004; Sekerina et al., 2018). For example, children may judge the sentence *Every bunny is in a box* as false when extra boxes or unrelated objects are present, despite the statement's logical truth. This phenomenon is typically observed in *extra object* contexts, as illustrated in Figure 1. The terminology used to describe children's Q-spreading errors has varied considerably across studies. For example, Roeper et al. (2004) distinguished between *bunny-spreading* and *classic Q-spreading* errors, whereas Freeman (1985) differentiated between *under-exhaustive* and *over-exhaustive* errors, noting that the latter correspond to Q-spreading errors. In the present study, the term *Q-spreading error* is used to refer to *over-exhaustive* errors, whereas *bunny-spreading error* denotes *under-exhaustive* errors.

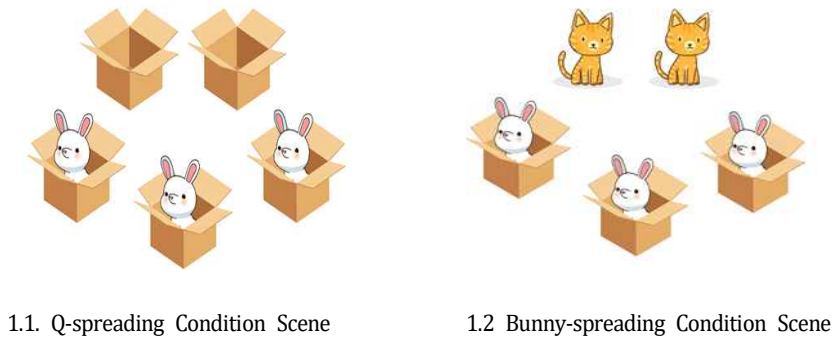


FIGURE 1
Sample Visual Scenes Illustrating the Q-Spreading (Left) and Bunny-Spreading (Right) Conditions

As shown in Figure 1.1, three bunnies are each placed in a box, with additional empty boxes remaining. The sentence *Every bunny is in a box* is logically true;

however, children often respond “false,” extending the quantifier’s scope from the set of bunnies to the empty boxes. This pattern constitutes a Q-spreading error. In Figure 1.2, each of the three bunnies is placed in a box, while unrelated extra objects (e.g., cats) remain. Although the sentence *Every bunny is in a box* is also true, children sometimes reject it as false. This type of error is classified as *bunny-spreading* (Roeper et al., 2004).

Bunny-spreading (under-exhaustive) errors tend to decline rapidly during early childhood and have therefore received comparatively less scholarly attention (Freeman, 1985; Roeper et al., 2004; Sekerina et al., 2018). In contrast, Q-spreading errors have been consistently observed among school-age children across a wide range of languages, including English, Dutch, Japanese, Russian, and Korean (Bucci, 1978; de Koster et al., 2018; Kang, 2001; Minai et al., 2012; Sekerina et al., 2018; Sekerina & Sauermann, 2017). Such cross-linguistic evidence suggests that Q-spreading errors represent a universal developmental phenomenon in language acquisition.

Importantly, Q-spreading errors are not restricted to children. Previous studies have shown that adults also exhibit these errors at notable rates, 24% in Aravind et al. (2017) and 25% in Brooks and Sekerina (2005). These findings suggest that Q-spreading errors cannot be attributed solely to immature language development; rather, they likely reflect more fundamental cognitive constraints on quantifier-scope processing.

Building on this body of research, the present study investigates how Korean-speaking children aged 5 to 6 process universally quantified sentences in real time. This age group represents a critical developmental period: children at this stage possess sufficient linguistic competence to perform comprehension tasks but still exhibit unstable control over quantifier interpretation, making them especially susceptible to Q-spreading errors (Minai et al., 2012; Sekerina & Sauermann, 2017). Despite extensive cross-linguistic research, little is known about how Korean children integrate linguistic and visual information when interpreting sentences with universal quantifiers. Previous studies have relied primarily on offline judgment tasks, which reveal only final responses and cannot capture the real-time processes leading to those errors. To address this limitation, the present study employs the visual-world eye-tracking paradigm, which enables continuous monitoring of visual attention during sentence comprehension. This approach allows for a fine-grained examination of the interaction between linguistic processing and visual attention, offering new insight into the mechanisms underlying children’s Q-spreading errors in Korean.

II. LITERATURE REVIEW

Children's difficulties in interpreting the universal quantifier *every* have been explained from three major perspectives: (i) faulty logical reasoning, (ii) partial linguistic competence, and (iii) full linguistic competence accompanied by processing limitations. From a cognitive-developmental perspective, Inhelder and Piaget (1964) viewed Q-spreading errors as a reflection of immature logical reasoning. According to this account, children's non-adult-like interpretations emerge from the gradual development of domain-general cognitive structures rather than from deficiencies in linguistic knowledge. Similarly, within a functionalist framework, Brooks and Braine (1996) proposed that children initially assume distributive quantifiers such as *each* and *every* enforce a strict one-to-one correspondence between individuals and objects. Thus, children associate quantification with the physical act of pairing rather than with the semantic properties of quantified noun phrases.

In contrast, generative approaches have attributed Q-spreading to non-adult-like grammatical representations. Philip (1995), in his *Event Quantification Hypothesis*, argued that children interpret universal quantifiers as ranging over events rather than over individuals, leading to over-exhaustive interpretations. This proposal has been further developed in subsequent studies that posit alternative syntax-semantics mappings, including weak quantification (Drozdz & van Loosbroek, 1998), noncanonical syntax-semantics mapping (Geurts, 2003), and syntactic restructuring analyses (Roeper, 2007; Roeper et al., 2004). In summary, these perspectives suggest that Q-spreading errors reflect temporary grammatical misanalyses rather than broader cognitive immaturity.

A third perspective emphasizes processing constraints. Poeppel and Wexler (1993), in their *full competence hypothesis*, argued that children possess adult-like knowledge of universal quantifiers but fail in certain experimental tasks due to processing limitations. Children's errors, in this view, arise not from incomplete grammar but from pragmatic infelicity, attentional constraints, and task-related cognitive load. Consistent with this interpretation, Crain et al. (1996), Meroni et al. (2007), and Rakhlin (2007) demonstrated that Q-spreading errors are more frequent when felicity conditions are violated, for instance, in yes/no judgment tasks lacking sufficient contextual support. Crain et al. (1996) further argued that children's apparent problems with universal quantification are largely pragmatic, stemming from experimental designs that fail to provide adequate contextual cues.

Expanding on this view, Rakhlin (2007) showed that children's responses vary

systematically with visual salience and contextual richness. Specifically, their judgments become more adult-like when the context is clearer or linguistically elaborated. Rakhlin's analysis, grounded in formal semantic theory, contends that children possess full adult-like semantics early in development but are still acquiring the pragmatic skills necessary to resolve ambiguity in minimal contexts. Similarly, studies by Sugisaki and Isobe (2001) and Minai et al. (2012) found that increasing the number of extraneous objects in experimental scenes reduces Q-spreading errors by diminishing the salience of individual distractors. When multiple extra objects are present, the uniqueness of each distractor is weakened, leading to fewer symmetrical, non-adult responses.

In addition, more recent research emphasizes the importance of processing and attentional factors in Q-spreading errors. For example, O'Grady et al. (2010) found that Japanese-speaking 5-year-olds made significantly fewer errors in an act-out task, where they manipulated physical cutouts, than in traditional picture-verification tasks. This finding suggests that visual distractions contribute substantially to Q-spreading errors, although the authors cautioned that act-out tasks might underestimate error rates by encouraging one-to-one pairing biases.

Further support for processing-based accounts comes from eye-tracking studies, which have provided valuable insights into real-time sentence comprehension. Using the Visual World Paradigm (Trueswell & Tanenhaus, 2005), Minai et al. (2012) found that Japanese-speaking preschoolers who produced Q-spreading errors fixated more frequently on irrelevant objects during and after sentence presentation. Moreover, these children performed poorly on executive function tasks such as the Dimensional Change Card Sort (DCCS), suggesting that underdeveloped attentional control and inhibitory mechanisms contribute to their errors. Consistent with findings on perseveration and attentional inertia (Diamond & Kirkham, 2005), Minai et al. concluded that children's difficulty in disengaging from salient distractors reflects the so-called "kindergarten-path effect" (Trueswell et al., 1999), in which young learners struggle to revise initial parsing commitments. Consistent with these findings, Woodard et al. (2016) emphasized that executive function plays a crucial role in overcoming these processing difficulties.

Building on processing-based accounts, we specifically integrate the role of executive control into our theoretical framework. Successful quantifier interpretation requires the real-time coordination of linguistic input with the visual scene, a process heavily reliant on general cognitive abilities (Miyake et al., 2000). The Q-spreading errors observed in visually complex scenes are hypothesized to

result from a failure in inhibitory control, a core component of executive function. This failure to suppress the strong, misleading attentional pull of the irrelevant 'Extra' object is consistent with the processing limitations noted in previous developmental studies (Trueswell et al., 1999). Moreover, the mechanism such as filtering irrelevant, salient information is a central aspect of cognitive control known to affect language processing demands (e.g., Novick et al., 2005). Thus, our eye-tracking analysis tests whether fixations on non-referential visual regions reflect a breakdown in inhibitory control.

Similarly, Sekerina and Sauermann (2017) found that Russian-speaking children frequently made Q-spreading errors by applying the quantifier to irrelevant items not mentioned in the sentence. Eye-tracking data revealed that such errors were accompanied by increased fixations on visually salient but unrelated objects, indicating that visual attention strongly influences quantifier interpretation. Extending this line of research, Sekerina et al. (2018) showed that English-speaking children aged 5 to 12 exhibited fixation patterns on distractors that reliably predicted erroneous responses, independent of verbal or nonverbal intelligence. Taken together, these findings suggest that Q-spreading errors result from the interaction among visual complexity, task demands, and domain-general cognitive constraints rather than from incomplete linguistic competence.

While most research on Q-spreading errors has focused on English-speaking children, studies on Korean-speaking children remain limited. To date, Kang (2001) has conducted the only systematic investigation of Korean children's comprehension of universal quantifiers (*every, each, all*) using a yes/no judgment task. Both English- and Korean-speaking children displayed high error rates (71% and 65%, respectively), confirming the cross-linguistic robustness of Q-spreading errors. Interestingly, left-spreading errors (object-quantifier, e.g., "Is a bear holding every honeypot?") occurred more frequently than right-spreading errors (subject-quantifier, e.g., "Is every bear holding a honeypot?"). Furthermore, syntactic complexity modulated performance: intransitives with prepositional phrases yielded the highest error rates, whereas transitives produced fewer errors. These results indicate that syntactic position and structural complexity influence children's susceptibility to Q-spreading errors.

In summary, Kang's (2001) study provided valuable cross-linguistic evidence that Korean children, like English-speaking counterparts, frequently exhibit Q-spreading errors. However, its reliance on offline yes/no judgment tasks limits interpretability, because such methods capture only the final outcome rather than the incremental processes underlying comprehension. In addition, the study did not

consider cognitive factors such as working memory or attentional control, which more recent eye-tracking studies have shown to play a crucial role in children's susceptibility to Q-spreading (Minai et al., 2012; Sekerina et al., 2018; Sekerina & Sauermann 2017). To address these limitations, the present study employs eye-tracking to examine the real-time processing of universally quantified sentences in Korean-speaking children aged 5 to 6. By integrating online measures of visual attention, this study aims to provide a more comprehensive account of the cognitive mechanisms underlying Q-spreading errors in Korean.

This study addresses the following research questions:

- 1) How do Age Group (Children vs. Adults) and Quantifier Condition (Q-spreading, Bunny, Control) interact to predict response accuracy in sentence-picture verification, and does Verb Type (Intransitive vs. Transitive) modulate these offline judgments?
- 2) How is real-time visual attention, measured by fixation preferences for Referential (Key) versus Non-referential (Extra) AOIs, modulated by Age Group and Quantifier Condition across the time course of the sentence (IOIs)?

III. METHOD

1. Participants

A total of 70 Korean children aged between 61 and 83 months (approximately 5–6 years) participated in the experiment. Data from 13 children were excluded due to excessive distraction or inability to complete the task, resulting in a final sample of 57 children ($M = 72.16$ months, $SD = 6.44$; 32 girls) included in the analyses. Among them, 25 were 5-year-olds ($M = 65.88$ months, $SD = 3.38$; 12 girls) and 32 were 6-year-olds ($M = 77.06$ months, $SD = 3.09$; 20 girls). All children were monolingual Korean speakers with normal or corrected-to-normal vision and hearing. Their vocabulary scaled scores on the Korean Wechsler Preschool and Primary Scale of Intelligence–Fourth Edition (K-WPPSI-IV) averaged 18.07 ($SD = 5.88$), which falls within the normal range. A total of 97 adults ($M = 22.42$ years, $SD = 2.82$; 58 females) participated as a comparison group. Adult participants were undergraduate and graduate students recruited from a regional

university and participated voluntarily. All participants reported normal or corrected-to-normal vision and hearing and were native speakers of Korean.

2. Experimental Task

1) Sentence-Picture Verification Task

The experimental materials were based on the sentence-picture verification paradigm widely used in Q-spreading research (e.g., Minai et al., 2012; Sekerina et al., 2018). Target sentences were designed to test whether children's interpretation of universally quantified sentences is influenced by visual context, which differed across three conditions: Q-spreading, Bunny-spreading, and Control. In addition, to examine whether verb type affects quantifier interpretation, the verb construction of the target sentences was manipulated. Previous studies, however, have typically focused on only one of these verb types. For example, Minai et al. (2012) included only transitive verbs, whereas Sekerina et al. (2018) examined intransitive verbs exclusively. To extend these findings, the present study investigates whether differences in verb transitivity influence Q-spreading errors.

Two syntactic forms were used: an intransitive/locative construction (IV), in which the predicate contained a prepositional phrase indicating location (as in Sekerina et al., 2018), as illustrated in (1); and a transitive construction (TV), in which the universal quantifier modified the subject and the predicate included a direct object (as in Minai et al., 2012), as illustrated in (2). This manipulation allowed us to examine whether verb transitivity (IV vs. TV) modulates Q-spreading tendencies.¹⁾

(1) Intransitive/Locative Construction (IV) in Korean

Modeun tokki-ga sangja an-ey isseoyo.
every bunny-NOM box in-LOC be-DECL
'Every bunny is in a box.'

(2) Transitive Construction (TV) in Korean

Modeun keobugi-ga usan-eul teul-go isseoyo.
every turtle-NOM umbrella-ACC hold-PROG be-DECL
'Every turtle is holding an umbrella.'

1) We use the following abbreviations in glossing our examples: ACC = accusative, DECL = declarative, NOM = nominative, PROG = progressive, LOC = locative.

The experiment employed a 3×2 factorial design with Quantifier Condition (Quantifier-spreading [Q-spreading], Bunny-spreading [Bunny], Control) and Verb Type (Intransitive/Locative [IV] vs. Transitive [TV]) as within-subject factors, yielding six experimental conditions. A total of 24 experimental trials were constructed, comprising eight items per condition. In the Q-spreading and Bunny-spreading conditions, the sentence-picture pairs were congruent, and the correct response was “yes.” In contrast, the Control condition presented incongruent sentence–picture pairs, for which the correct response was “no.” This design produced 16 “yes” trials and 8 “no” trials across the experimental set. Representative examples of sentence-picture pairings for each condition are illustrated in Figure 2.

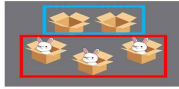
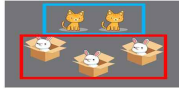
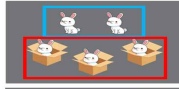
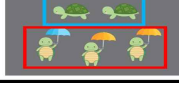
Condition	Verb Type	Example sentence	Example picture
Q-spreading	IV	Every bunny is in a box (correct = ‘yes’)	
	TV	Every turtle is holding an umbrella. (correct = ‘yes’)	
Bunny-spreading	IV	Every bunny is in a box (correct = ‘yes’)	
	TV	Every turtle is holding an umbrella. (correct = ‘yes’)	
Control	IV	Every bunny is in a box (correct = ‘no’)	
	TV	Every turtle is holding an umbrella. (correct = ‘no’)	

FIGURE 2

Examples of Sentence–Picture Verification Task Stimuli across Quantifier Conditions and Verb Types (Blue Boxes: Extra Objects, Red Boxes: Key Objects)

In addition, 40 filler trials were included to increase variability and reduce predictability. Among them, 16 fillers were composed of active and passive voice constructions paired with images designed to yield an equal number of “yes” and “no” responses (8 each). The remaining 24 fillers were semantically neutral distractors, designed to require “yes” responses in 8 trials and “no” responses in

16 trials. This composition ensured a balanced overall response distribution, with 32 “yes” and 32 “no” trials across the entire experiment. All sentences were recorded by a female native speaker of Korean at a natural speaking rate. Further details regarding stimulus presentation and response procedures are provided in Section 3.4.

2) Eye-Tracking Task

Eye movements were recorded throughout the sentence-picture verification task using a Tobii Pro X3-120 remote eye-tracking system (Tobii Technology), operating at a sampling rate of 120Hz. Visual stimuli were displayed on a 1920×1080 resolution monitor, and auditory sentence stimuli were delivered through headphones. Participants were seated approximately 60-70cm from the display and were instructed to minimize head movement. A chin-and-forehead rest was used to maintain head stability and ensure consistent viewing geometry. Before the main task, a standard 9-point calibration procedure was administered, followed by validation to confirm accurate and reliable gaze tracking.

Fixations were defined as gaze positions that remained within 0.5 degrees of visual angle for at least 100 milliseconds. This threshold ensured that only meaningful visual attention was analyzed. During each trial, participants initially viewed a centrally presented fixation cross while hearing a preparatory prompt (“Ready?”). Once gaze was confirmed on the fixation point, the target image appeared on the screen, and the corresponding sentence was played. After the auditory stimulus ended, a response screen was displayed prompting the participant to judge whether the sentence matched the picture by pressing the appropriate key (“Yes” or “No”) on a response box. The entire sentence-picture task, including calibration and trials, took approximately 20 minutes to complete. Figure 3 provides a schematic overview of the trial sequence implemented in the eye-tracking session.

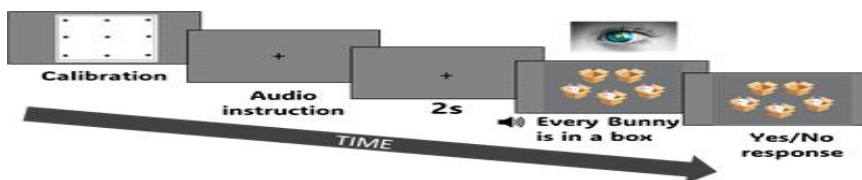


FIGURE 3

Schematic of the Eye-Tracking Procedure during the Sentence-Picture Verification Task

3. Procedure

All participants completed the experimental tasks individually in a quiet room. Prior to participation, written informed consent was obtained from adult participants, while children took part with the written consent of their parents. After completing a brief demographic questionnaire, participants were given verbal instructions about the sentence-picture verification task and completed several practice trials before proceeding to the experimental session. All procedures were approved by the Institutional Review Board of XX University (IRB No. 202107-SB-168-01).

4. Statistical Analyses

To investigate how children and adults interpret universal quantifier sentences, we analyzed both response accuracy and eye-tracking measures using mixed-effects modeling approaches. The analysis was based on the 3 (Quantifier Condition: Q-spreading, Bunny, Control) $\times$ 2 (Verb Type: IV, TV) factorial design, as described in Section 3.2 (11-12 Experimental Tasks). For both data types, fixed effects included Age Group (Children, Adults), Quantifier Condition, and Verb Type, along with all relevant interactions. All models included random intercepts for Participant and Item. Post-hoc comparisons were conducted using estimated marginal means (EMMs) with Tukey correction for multiple comparisons (Lenth, 2025), applied to both accuracy and eye-tracking outcomes as appropriate. All analyses were conducted in R using the lme4, lmerTest, and emmeans packages (Bates et al., 2015; Kuznetsova et al., 2017).

Response accuracy in the sentence-picture verification task was analyzed using linear mixed-effects models (LMMs) with Gaussian errors and an identity link, treating accuracy as a percentage outcome. This analysis assessed whether participants correctly judged the match between the auditory quantifier sentences and the corresponding visual scenes, and whether accuracy varied as a function of Age Group, Quantifier Condition, and Verb Type. Although a trial-level Binary Logistic Mixed Model (BLMM) is the conventional approach for binary outcomes, we used an aggregated LMM because the severe ceiling effect in the adult data introduced distortions and reduced statistical power in the BLMM due to the compressive nature of the logit link. Under these conditions, the aggregated LMM provided a more sensitive and reliable means of testing our core hypothesis regarding the Age Group $\times$ Quantifier Condition interaction. The primary model

included fixed effects of Age Group, Quantifier Condition, Verb Type, and their interactions, along with a random intercept for Participant. Models were estimated using maximum likelihood, and t-tests were computed using Satterthwaite's approximation for degrees of freedom. Where appropriate, Type III *F*-tests were reported. Estimated marginal means (EMMs) and post hoc pairwise comparisons were obtained using the emmeans package, reported as percentage-point estimates, with Kenward-Roger degrees of freedom and Tukey adjustments for multiple comparisons. For the exploratory analysis within the child group (5- vs. 6-year-olds), the same model structure was applied, with Age Group replaced by Child Age as a fixed factor.

Eye-tracking data were analyzed using LMMs applied to area-normalized fixation duration ratios for two predefined Areas of Interests (AOIs): the Key object (referential target) and the Extra object (non-referential distractor). Fixation durations were normalized by each AOI's relative screen area to control for differences in visual size (Bindemann et al., 2009; Hills & Pake, 2013), with values above 1.0 indicating preferential attention beyond chance level. Condition-level effects were visualized using EMMs derived from the LMMs, representing model-based averages of these area-normalized fixation ratios.

To capture the time course of referential attention, each sentence was segmented into four Intervals of Interests (IOIs) corresponding to linguistically meaningful regions of the sentence: IOI 1 (subject noun phrase), IOI 2 (locative or accusative phrase), IOI 3 (verb), and IOI 4 (post-verbal silence). As shown in Figure 3, IOIs were aligned with the auditory sentence structure, while AOIs were spatially defined on the image display to isolate attention to referential (Key) versus non-referential (Extra) objects. The LMMs for eye-tracking data included fixed effects for AOI (Key vs. Extra) and IOI (1-4), and their interactions with Age Group, Quantifier Condition, and Verb Type.

IV. RESULTS

1. Accuracy

The LMM analysis revealed a significant main effect of Age Group, $\beta = -28.59$, $SE = 4.16$, $t = -6.88$, $p < .001$, with adults performing more accurately than children across conditions (EMMs: Adults = 97.9%, Children = 54.7%). Quantifier Condition also showed a significant main effect, $F(2, 739.45) = 34.22$, $p < .001$. Post hoc

EMM comparisons revealed significant accuracy differences among all three conditions: Control vs. Bunny (difference = 5.88%, $SE = 1.96$, $t = 3.00$, $p = .008$), Control vs. Q-spreading (difference = 13.05%, $SE = 1.96$, $t = 6.68$, $p < .001$), and Bunny vs. Q-spreading (difference = 7.18%, $SE = 1.95$, $t = 3.69$, $p < .001$). No significant main or interaction effects involving Verb Type were observed.

TABLE 1

Estimated Marginal Means (EMMs) of Accuracy (%) by Age Group and Quantifier Condition

Age Group	Quantifier Condition	EMM (%)	SE
Adults	Q-spreading	96.8	1.87
	Bunny	98.5	1.88
	Control	98.3	1.88
Children	Q-spreading	38.9	2.37
	Bunny	55.0	2.35
	Control	70.4	2.41

A significant Age Group $\times$ Quantifier Condition interaction emerged ($F(2, 739.45) = 28.22$, $p < .001$). Among adults, accuracy remained near ceiling across all conditions ($EMM = 97.9\%$, 95% CI [95.5, 100.3]), and pairwise comparisons showed no reliable differences (all $ps > .05$). In contrast, children exhibited significant condition effects: accuracy in the Control condition was significantly higher than in the Bunny (difference = 15.46%, $SE = 3.16$, $t = 4.90$, $p < .001$) and Q-spreading conditions (difference = 31.53%, $SE = 3.17$, $t = 9.96$, $p < .001$), and performance in Bunny exceeded that in Q-spreading (difference = 16.07%, $SE = 3.12$, $t = 5.14$, $p < .001$; see Figure 4 and Table 1). As shown in Figure 4, adults demonstrated uniformly high performance, whereas children displayed a graded decline from Control to Q-spreading, indicating increasing difficulty with quantifier interpretation.

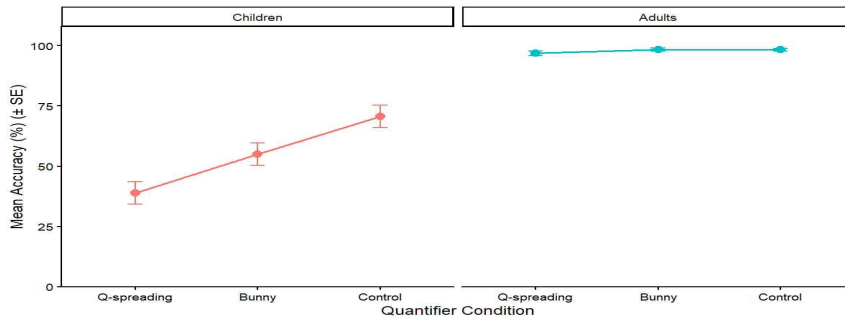


FIGURE 4

Mean Accuracy ($\pm$ SE; estimated marginal means from the LMM) by Age Group (Children, Adults) and Quantifier Condition (Q-spreading, Bunny, Control)

To further examine developmental differences within the child group, a follow-up LMM analysis was conducted comparing 5- and 6-year-olds, including fixed effects of Child Age, Quantifier Condition, Verb Type, and their interactions. No significant main or interaction effects involving Child Age were found, indicating that overall accuracy was comparable between 5- and 6-year-olds. Similarly, Verb Type did not show any significant main or interaction effects. In contrast, Quantifier Condition exhibited a significant main effect, with accuracy following the same graded pattern observed in the full sample: Control > Bunny > Q-spreading. These results suggest that children's performance differences were driven primarily by sentence condition rather than age, and that difficulties in quantifier processing were evident across both age groups. All results from this analysis are summarized in Table 2.

TABLE 2

Estimated Marginal Means (EMMs) of Accuracy by Child Age and Quantifier Condition

Age Group	Quantifier Condition	EMM (%)	SE
5-year-old	Q-spreading	40.4	5.44
	Bunny	47.5	5.44
	Control	63.2	5.44
6-year-old	Q-spreading	37.7	4.81
	Bunny	60.7	4.77
	Control	76.6	4.97

2. Eye Movement Patterns

Eye-tracking data were analyzed using LMMs on area-normalized fixation duration ratios. Fixed effects included AOI (Key vs. Extra), IOI (1-4), Age Group (Children, Adults), Quantifier Condition (Control, Bunny, Q-spreading), and Verb Type (IV, TV), along with all relevant interactions. Random intercepts for Participant and Item were included. Eye-tracking data from 4 children and 13 adults were excluded due to calibration failure or excessive data loss. The final analyses therefore comprised valid gaze data from 53 children and 84 adults. All results are reported as EMMs derived from the LMMs. Full fixed-effect estimates are provided in Appendix A, and the present section summarizes key effects.

1) Age Group Differences in Referential Attention Across Quantifier Conditions

The analysis revealed a significant three-way interaction between Age Group, Quantifier Condition, and AOI ($p < .001$). To interpret this effect, post-hoc comparisons were conducted using EMMs computed from the LMM, with multiple comparison adjustments (Tukey for main effects and Holm-Bonferroni for interaction effects). All comparisons were conducted using the emmeans package in R (Lenth, 2025).

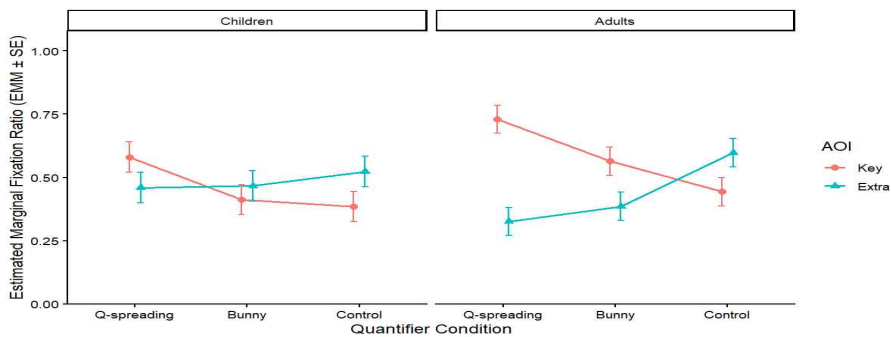


FIGURE 5

Estimated Marginal Means (EMMs) of Area-Normalized Fixation Ratios to Key and Extra AOIs by Quantifier Condition and Age Group

Figure 5 illustrates the Age Group × Quantifier Condition × AOI interaction. Among adults (right panel), fixation ratios clearly favored the Key AOI in Bunny

and Q-spreading conditions ($ps < .001$), but reversed in Control, where the Extra AOI was preferred. In contrast, children (left panel) showed more variable gaze patterns across conditions. In Control, they preferred the Extra AOI ($p < .001$), consistent with adults. In Bunny, no reliable AOI preference was observed ($p = .076$), whereas in Q-spreading, they showed a significant Key-over-Extra preference ($p < .001$). This pattern suggests that whereas adults consistently modulated visual attention to referential targets according to condition structure, children's referential gaze allocation was more condition-dependent, with strong Key fixation emerging only under quantifier load.

2) Modulations by Time Course and Verb Type

While the Age Group $\times$ Quantifier Condition $\times$ AOI interaction captured the primary pattern of referential attention, this effect was further qualified by sentence timing and Verb Type, as indicated in significant four-way interactions (see Appendix A). Figure 6 plots the EMMs across IOIs for each Age Group and Quantifier Condition, illustrating the four-way interaction (Age Group $\times$ Quantifier Condition $\times$ AOI $\times$ IOI).

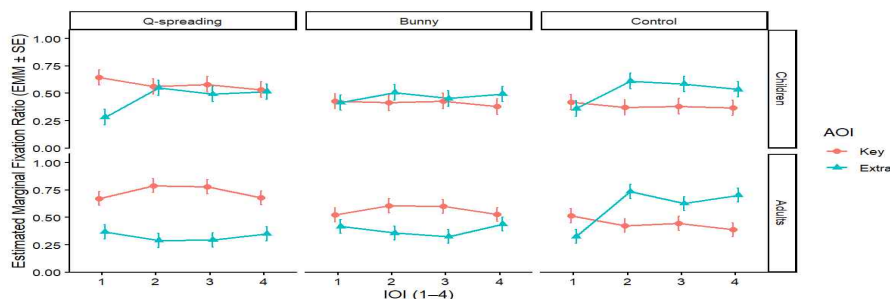


FIGURE 6

EMMs across IOIs (1-4) for Key and Extra AOIs, Plotted Separately by Quantifier Condition and Age Group

The time-resolved patterns revealed distinct developmental trajectories. In the Q-spreading condition, children showed a brief Key-over-Extra preference confined to IOI 1, with no reliable AOI difference at IOIs 2-4 ($ps > .05$), whereas adults maintained a strong Key preference across all IOIs ($ps < .01$). In the Control condition, both groups exhibited an Extra-over-Key pattern from IOI 2 onward,

although adults displayed an early Key-over-Extra peak at IOI 1 before reversing in later intervals. In the Bunny condition, adults demonstrated a Key advantage primarily at IOIs 2-3 ($ps < .01$), whereas children showed no reliable AOI preference across IOIs ($ps > .05$). Taken together, these findings indicate that referential gaze in adults remains stable and tightly aligned with condition structure over time, whereas children's fixations are more transient, emerging briefly at the beginning of the sentence and further modulated by verb type and visual competition.

Appendix A presents additional effects of Verb Type. In the Q-spreading condition, children exhibited a stronger Key preference with IV verbs, whereas their AOI preference was absent with TV verbs. Adults, by contrast, maintained a consistent Key preference regardless of verb type. Thus, these findings indicate that time course and verb type modulate the strength but not the direction of referential attention, particularly among children.

Overall, adults demonstrated selective and condition-sensitive gaze behavior, consistently fixating on Key referents in the Bunny and Q-spreading conditions, while shifting toward Extra referents in Control during later IOIs. Children, by contrast, showed more variable patterns of referential attention, with a brief Key-over-Extra preference early in Q-spreading and no reliable AOI preference in Bunny. These differences unfolded dynamically over the course of the sentence, typically emerging around IOIs 2-3, and were further modulated by Verb Type. However, such modulations were secondary to the core Age Group $\times$ Quantifier Condition $\times$ AOI interaction.

Analyses of Fixation Count(FC) and Total Fixation Duration(TFD) measures yielded consistent AOI patterns, supporting the robustness of the referential attention effects. A full summary of TFD and FC comparisons is provided in Appendix A and B. The only divergence appeared at the global two-way level: FC, but not TFD, showed a significant Age Group $\times$ Quantifier Condition interaction (FC: $F = 3.37$, $p = .034$; TFD: $F = 2.29$, $p = .101$), driven by a greater number of fixations among adults than children in the Control condition. AOI-specific patterns remained consistent across measures.

3) Accuracy-Related Modulations in Children's Gaze

To further examine how accuracy influences referential attention, children's gaze patterns were analyzed as a function of response correctness. A follow-up LMM analysis was conducted on children's trials with valid eye-tracking and accuracy

data. The model included Response (Correct vs. Error), AOI, and Quantifier Condition as fixed effects, along with all interactions. Participant and Item were modeled as random intercepts, and Vocabulary Score was included as a covariate. The vocabulary score was applied only in the child analysis to control for variation in language ability that might otherwise influence sentence-picture verification performance and gaze behavior. Because adults' vocabulary scores were near ceiling, no such covariate was included in the adult analyses. Full model estimates are reported in Appendix C.

The analysis revealed a significant Response $\times$ AOI interaction ($\beta = -0.17$, $SE = 0.07$, $t = -2.51$, $p = .012$), indicating that gaze allocation differed between Correct and Error trials. As shown in Figure 7, children displayed a marked Key-over-Extra preference in the Q-spreading condition on Correct trials ($p < .001$), which was diminished to near parity on Error trials. In the Control condition, Correct trials showed an Extra-over-Key preference ($p < .001$), which was similarly attenuated on Error trials. In contrast, no consistent AOI preference was observed in Bunny, regardless of trial accuracy (Response $\times$ AOI $\times$ Bunny: $t = 1.04$, $p = .299$).

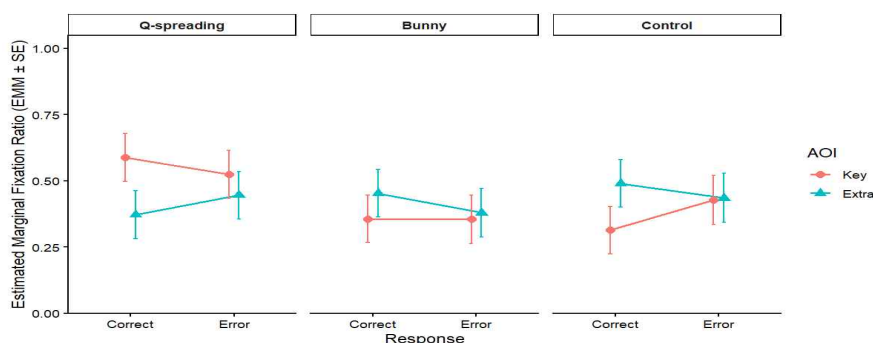


FIGURE 7

EMMs to Key and Extra AOIs by Verb Type (IV vs. TV), Age Group, and Quantifier Condition (collapsed across IOIs)

A significant three-way interaction of Response $\times$ AOI $\times$ Quantifier Condition ($\beta = 0.31$, $SE = 0.09$, $t = 3.31$, $p = .001$) confirmed that the reduction of Key preference on Error trials was condition-specific, most prominent in Q-spreading. Importantly, on Correct trials, children did not maintain a uniform Key preference across conditions; rather, this preference was selectively observed in Q-spreading. These findings suggest that accurate quantifier interpretation in children is closely

linked to referential fixation on the intended target. Conversely, Error trials showed more distributed or misaligned gaze patterns, particularly under increased referential load, when the sentence required integrating a universal quantifier with visually competing objects in the scene. This supports the interpretation that gaze directed toward referentially relevant targets (e.g., Key AOIs) functions as a behavioral marker of successful sentence-picture integration during development.

V. DISCUSSION

The present study examined how Korean-speaking children and adults interpret universally quantified sentences by integrating behavioral accuracy and eye-tracking data. Adults performed near ceiling, exhibiting minimal condition differences, whereas children showed systematic variation in both accuracy and gaze behavior as a function of visual context. Accuracy declined progressively from the Control to Bunny to Q-spreading conditions, indicating increasing difficulty in integrating universal quantifiers with visually competing referents. Although 5-year-olds showed slightly lower accuracy than 6-year-olds, this difference was not significant once random effects for Participant and Item were accounted for, suggesting that apparent age differences reflect individual variability rather than a robust developmental shift in quantifier comprehension. This pattern supports the view that children's interpretive errors arise from immature executive control and limited attentional flexibility rather than from deficits in semantic knowledge (Minai et al., 2012) and aligns with broader evidence that processing-based variability precedes age-graded stabilization of referential integration skills (Kidd & Donnelly, 2020; Siegler, 2006).

Nevertheless, overall accuracy among 5- to 6-year-olds remained at or near chance in the Bunny condition and dropped below chance in the Q-spreading condition, suggesting that the ability to compute universal quantification in visually competitive contexts is still developing at this age. Similar performance patterns have been reported for Russian-speaking preschoolers (43% accuracy; Sekerina & Sauermann, 2017) and Japanese children tested with single-extra-object displays (Minai et al., 2012), both of whom exhibited substantial interference from visually salient but truth-conditionally irrelevant objects. Likewise, similar findings for Korean children's sentence-picture verification tasks (Choi, 2005) suggest that this difficulty reflects a broader developmental constraint rather than a language-specific phenomenon. While accuracy data alone cannot distinguish

between processing limitations and incomplete semantics, the cross-linguistic convergence of error patterns suggests a shared vulnerability to visual competition during early quantifier comprehension.

We further position the Korean findings within the broader cross-linguistic literature on quantifier interpretation. Korean-speaking children exhibit performance patterns that closely resemble those reported for Japanese-speaking children (Minai et al., 2012), Russian-speaking children (Sekerina & Sauermann, 2017), and English-speaking children (Sekerina et al., 2018), particularly in their susceptibility to visual interference and their reliance on executive control to resolve referential competition. Notably, these parallels arise despite substantial typological differences among the languages, including contrasts in word order (SOV vs. SVO), case-marking systems, verb-final structure, and quantifier morphology, indicating that Q-spreading errors are driven by domain-general processing and developmental constraints rather than language-specific grammatical properties. By integrating Korean-specific findings with evidence from Japanese, Russian, and English, the present study demonstrates that Korean follows a broader cross-linguistic developmental trajectory in quantifier acquisition. This integrated perspective highlights the convergence of processing-based and pragmatic accounts across languages while identifying promising directions for future typological comparison.

Eye-tracking data clarified the mechanism underlying these errors. Adults consistently directed fixation toward referentially relevant targets (Key AOIs) in the Bunny and Q-spreading conditions, whereas children's gaze patterns were less stable, showing greater attention to Extra AOIs in Control and only transient increases in Key fixations under Q-spreading. This pattern parallels the results of Sekerina and Sauermann (2017), who reported that increased fixations on Extra entities correlate with Q-spreading errors. The present data refine this processing-overload account by specifying the hierarchical relationship between attentional control and interference: limited attentional flexibility may function similarly to reduced executive control, amplifying interference from visually salient but non-referential objects. This interference, in turn, increases cognitive load precisely when a one-to-one mapping between agents and objects must be established—the temporal locus of spreading identified by Sekerina and Sauermann. As previously discussed, Minai et al. (2012) also found that lower scores on the DCCS, a measure of cognitive control, were associated with greater Q-spreading tendencies. This correspondence supports the view that visual competition observed in the present study is not an independent causal factor but an

observable manifestation of limited cognitive control during semantic integration. It therefore offers a real-time behavioral indicator of children's difficulty in managing referential mappings under heightened processing demands (Huettig & Mani, 2016).

Although Verb Type (transitive vs. intransitive/locative) did not produce significant effects in accuracy, eye-tracking patterns showed a slight modulation of referential attention across event types. Under intransitive verbs, children tended to sustain longer fixations on Key referents, whereas this preference weakened with transitive verbs, which explicitly encode agent-object relations. This pattern suggests that verb semantics influence how comprehenders distribute visual attention, consistent with evidence that verbs rapidly constrain visual anticipation during incremental sentence processing (Altmann & Kamide, 1999) and that event structure subtly shapes the allocation of visual attention across potential referents (Sekerina & Saueremann, 2017).

Beyond processing load, the present findings also intersect with Rakhlin's (2007) pragmatic account, which explains quantifier spreading as a consequence of children's difficulty in restricting the domain of indefinites and their overreliance on visual salience when contextual cues are weak. In this view, spreading emerges when a child interprets an indefinite as a singleton referent, for instance, treating *Every bunny is in a box* as bunnies sharing one box rather than each having its own. This interpretation makes the universal quantifier logically incorrect but pragmatically reasonable given the visual scene. Integrating this pragmatic perspective with the processing account suggests that limited cognitive control and pragmatic immaturity operate at complementary levels, while reduced executive control yields visual interference during referential mapping, immature pragmatic filtering prevents children from overriding visually salient but contextually inappropriate interpretations (Gelman et al., 2015; Rakhlin, 2007). Supporting this integrated view, the present eye-tracking evidence, that is, frequent fixations on Extra objects before sentence offset and diminished attention to Key referents on error trials, demonstrates that both processing and pragmatic constraints jointly shape the developmental trajectory of quantifier interpretation.

The present findings are also compatible with the full-competence account, which attributes Q-spreading errors to processing limitations rather than grammatical immaturity (Minai et al., 2012; Philip, 1995; Sekerina & Saueremann, 2017). At the same time, the persistent sensitivity to visual context observed in this study may reflect broader domain-general reasoning constraints, as noted in early cognitive accounts (Inhelder & Piaget, 1964). This perspective suggests that the development of quantifier comprehension involves a gradual coordination

between linguistic computation and nonlinguistic reasoning systems.

The alignment between gaze allocation and accuracy further supports the account proposed by Sekerina and Sauermann (2017), which claims that young comprehenders possess full semantic competence but experience greater processing demands when integrating linguistic and visual information. This difficulty is specifically linked to the developing capacities of executive control (Miyake et al., 2000), particularly inhibitory control. When visual distraction was effectively inhibited and fixation was directed toward the referential target, children's truth-value judgments closely matched those of adults. In contrast, when inhibition failed, reflecting a lapse in the cognitive control mechanisms necessary for filtering irrelevant context (Novick et al., 2005; Trueswell et al., 1999), gaze dispersion coincided with non-adult-like interpretations. This close correspondence between gaze and accuracy underscores that fixation on referentially relevant entities functions as a real-time behavioral indicator of successful quantifier integration and reflects the dynamic coordination of linguistic computation and visual context during development (Fernald et al., 2008).

Despite offering a detailed account of how children coordinate linguistic and visual information during quantifier interpretation, the present study has several methodological and developmental limitations. The sample size, while sufficient for detecting group-level effects, may not capture the full range of individual differences in executive control that influence processing efficiency. Expanding the participant pool to include a cognitively diverse sample and incorporating independent measures of cognitive flexibility, inhibitory control, and working memory would clarify how domain-general mechanisms contribute to Q-spreading susceptibility.

Additionally, the present study focused on preschool-aged children (5-6 years old), whose performance in the Q-spreading and Bunny conditions was at or below chance level. This finding indicates that the ability to reliably interpret universal quantifiers is still developing at this age. Future research should include early school-aged children to determine when accuracy and gaze patterns approximate adult performance, and whether this developmental transition coincides with improvements in executive control. A broader age comparison would help delineate the developmental timeline of quantifier comprehension more precisely.

Finally, although the limited verb set used in the present stimuli enhanced experimental control, it also restricted the generalizability of the findings. Systematically varying verb semantics (e.g., motion, stative, and locative predicates) and manipulating visual complexity or object salience could provide deeper insight

into how event interpretation interacts with quantifier processing. In addition, embedding quantified sentences within discourse or narrative contexts would allow for a more contextually grounded assessment of how children integrate linguistic, visual, and pragmatic information during comprehension. Cross-linguistic and typological comparisons, particularly among languages differing in quantifier morphology or argument structure, could further clarify whether the developmental trajectory observed in Korean reflects universal cognitive mechanisms or language-specific processing patterns.

VI. CONCLUSION

This study examined how Korean-speaking children and adults interpret universally quantified sentences using both behavioral accuracy and eye-tracking data. Adults performed near ceiling across all conditions, whereas 5- to 6-year-old children showed marked susceptibility to visual interference, particularly in the Q-spreading condition. Their gaze patterns revealed increased attention to visually salient but truth-conditionally irrelevant objects, indicating that Q-spreading errors primarily reflect attentional competition rather than a lack of semantic knowledge. These findings support processing-based accounts of quantifier interpretation, emphasizing the interaction between executive control and pragmatic reasoning in resolving referential ambiguity. Acquiring universal quantification entails not only understanding the semantics of *every* but also coordinating linguistic, visual, and cognitive processes in real time. Limited attentional control and developing pragmatic filtering together account for children's difficulty in restricting quantifier scope to contextually appropriate domains. The results favor an integrated framework linking linguistic competence with domain-general cognitive mechanisms. Eye-tracking provided a real-time indicator of children's integration processes, revealing when comprehension succeeds or fails. Q-spreading errors thus represent not a failure of understanding, but an intermediate stage in the gradual alignment of linguistic and cognitive systems during early language development. These results also carry important implications for language acquisition. The link between Q-spreading errors and failures to inhibit distractors suggests that acquiring universal quantification requires not only semantic understanding but also the maturation of inhibitory control. This insight extends to second language acquisition, as young L2 learners with still-developing executive functions may similarly depend on salient visual cues when processing complex quantified

expressions. Thus, the study underscores the role of domain-general cognitive development in both L1 and L2 quantifier interpretation.

REFERENCES

- Altmann, G. T., & Kamide, Y. (1999). Incremental interpretation at verbs: Restricting the domain of subsequent reference. *Cognition*, 73(3), 247–264. [https://doi.org/10.1016/S0010-0277\(99\)00059-1](https://doi.org/10.1016/S0010-0277(99)00059-1)
- Aravind, A., de Villiers, J., de Villiers, P. A., Lonigan, C. J., Phillips, B. M., Clancy, J., Landry, S. H., Swank, P. R., Assel, M., & Taylor, H. B. (2017). Children's quantification with *every* over time. *Glossa: A Journal of General Linguistics*, 2(1), Article 43, 1–16. <https://doi.org/10.5334/gjgl.166>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67, 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bindemann, M., Scheepers, C., & Burton, A. M. (2009). Viewpoint and center of gravity affect eye movements to human faces. *Journal of Vision*, 9(7), 1–16.
- Brooks, P. J., & Braine, M. D. (1996). What do children know about the universal quantifiers all and each? *Cognition*, 60(3), 235–268. [https://doi.org/10.1016/0010-0277\(96\)00712-3](https://doi.org/10.1016/0010-0277(96)00712-3)
- Brooks, P. J., & Sekerina, I. A. (2005). Shortcuts to quantifier interpretation in children and adults. *Language Acquisition*, 13, 177–206.
- Bucci, W. (1978). The interpretation of universal affirmative propositions. *Cognition*, 6(1), 55–77. [https://doi.org/10.1016/0010-0277\(78\)90009-4](https://doi.org/10.1016/0010-0277(78)90009-4)
- Choi, J. (2005). Structures and semantic interpretations of Korean universal quantifier construction. *HANGEUL*, 267, 89–118.
- Crain, S., Thornton, R., Boster, C., Conway, L., Lillo-Martin, D., & Woodams, E. (1996). Quantification without qualification. *Language Acquisition*, 5(2), 83–153. https://doi.org/10.1207/s15327817la0502_2
- Diamond, A., & Kirkham, N. (2005). Not quite as grown-up as we like to think: Parallels between cognition in childhood and adulthood. *Psychological Science*, 16(4), 291–297. <https://doi.org/10.1111/j.0956-7976.2005.01530>
- Droz, K., & van Loosbroek, E. (1998). Weak quantification, plausible dissent, and the development of children's pragmatic competence. In A. Greenhill, H. Littlefield & C. Tano (Eds.), *Proceedings of the 23rd Boston university conference on language development* (pp. 184–195). Cascadia Press.

- Fernald, A., Zangl, R., Portillo, A. L., & Marchman, V. A. (2008). Looking while listening: Using eye movements to monitor spoken language comprehension by infants and young children. In I. A. Sekerina, E. M. Fernández, & H. Clahsen (Eds.), *Developmental psycholinguistics: On-line methods in children's language processing* (pp. 97–135). John Benjamins.
- Freeman, N. H. (1985). Reasonable errors in basic reasoning. *Educational Psychology*, 5(3-4), 239–249. <https://doi.org/10.1080/0144341850050307>
- Gelman, S. A., Leslie, S.-J., Was, A. M., & Koch, C. M. (2015). Children's interpretations of general quantifiers, specific quantifiers and generics. *Language, Cognition and Neuroscience*, 30(4), 448–461.
- Geurts, B. (2003). Reasoning with quantifiers. *Cognition*, 86(3), 223–251. [https://doi.org/10.1016/S0010-0277\(02\)00180-4](https://doi.org/10.1016/S0010-0277(02)00180-4)
- Hills, P. J., & Pake, J. M. (2013). Eye-tracking the own-race bias in face recognition: Revealing the perceptual and socio-cognitive mechanisms. *Cognition*, 129(3), 586–597. <https://doi.org/10.1016/j.cognition.2013.08.012>
- Huetting, F., & Mani, N. (2016). Is prediction necessary to understand language? Probably not. *Language, Cognition and Neuroscience*, 31(1), 19–31.
- Inhelder, B., & Piaget, J. (1964). *The early growth of logic in the child: Classification and seriation* (1st ed.). Routledge & Paul.
- Kang, H.-K. (2001). Quantifier spreading: Linguistic and pragmatic considerations. *Lingua*, 111(8), 591–627. [https://doi.org/10.1016/S0024-3841\(00\)00042-5](https://doi.org/10.1016/S0024-3841(00)00042-5)
- Kidd, E., & Donnelly, S. (2020). Individual differences in first language acquisition. *Annual Review of Linguistics*, 6(1), 319–340. <https://doi.org/10.1146/annurev-linguistics-011619-030326>
- de Koster, A., Spenader, J., & Hendriks, P. (2018). Are children's overly distributive interpretations and spreading errors related? In A. B. Bertolini & M. J. Kaplan (Eds.), *Proceedings of the 42nd annual Boston university conference on language development* (pp. 413–426). Cascadilla Press.
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*, 82, 1–26. <https://doi.org/10.18637/jss.v082.i13>
- Lenth, R. V. (2025). Emmeans: Estimated marginal means, aka least-squares means (R package version 1.11.2-8). <https://rvlenth.github.io/emmeans/>
- Meroni, L., Gualmini, A., & Crain, S. (2007). The strength of the universal quantifier in child language. In A. Belikova, L. Meroni & M. Umeda (Eds.). *Proceedings of the 2nd conference on generative approaches to language acquisition North America (GALANA)* (pp. 277–284). Cascadilla Press.

- Minai, U., Jincho, N., Yamane, N., & Mazuka, R. (2012). What hinders child semantic computation: children's universal quantification and the development of cognitive control. *Journal of Child Language*, 39(5), 919–956. <https://doi.org/10.1017/S0305000911000316>
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex “frontal lobe” tasks: A latent variable analysis. *Cognitive Psychology*, 41(1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>
- Novick, J. M., Trueswell, J. C., & Thompson-Schill, S. L. (2005). Cognitive control and parsing: Reexamining the role of Broca's area in sentence comprehension. *Cognitive, Affective, and Behavioral Neuroscience*, 5(3), 263–281.
- O'Grady, W., Suzuki, T., & Yoshinaga, N. (2010). Quantifier spreading: New evidence from Japanese. *Language Learning and Development*, 6(2), 116–125. <https://doi.org/10.1080/15475440903352799>
- Philip, W. (1995). *Event quantification in the acquisition of universal quantification*. (Unpublished doctoral dissertation). University of Massachusetts Amherst.
- Poeppel, D., & Wexler, K. (1993). The full competence hypothesis of clause structure in early German. *Language*, 69(1), 1–33.
- Rakhlin, N. (2007). A new pragmatic account of quantifier-spreading. *Nanzan Linguistics*, 8, 239–282.
- Roeper, T. (2007). The maximization of falsifiability: How to acquire the logic of implicatures from the illogic experience. In Y. Otsu (Ed.), *Proceedings of the 8th Tokyo conference on psycholinguistics* (pp. 1–19). Hituzi Syobo.
- Roeper, T., Strauss, U., & Pearson, B. Z. (2004). *The acquisition path of quantifiers: Two kinds of spreading*. Unpublished manuscript, University of Massachusetts at Amherst. http://people.umass.edu/roeper/online_papers/pathof%20acquisition3-15c.pdf
- Sekerina, I. A., Brooks, P. J., Campanelli, L., & Schwartz, A. M. (2018). Quantifier spreading in school-age children: An eye-tracking study. In K. É. Kiss & T. Zétényi (Eds.), *Linguistic and cognitive aspects of quantification* (pp. 171–192). Springer. https://doi.org/10.1007/978-3-319-91566-1_8
- Sekerina, I., & Sauermann, A. (2017). Quantifier spreading in child eye movements: A case of Russian quantifier kazhdyi ‘every’. *Glossa: A Journal of general linguistics*, 2(1). 1–18. <https://doi.org/10.5334/gjgl.109>
- Siegler, R. S. (2006). Microgenetic analyses of learning. In D. Kuhn, R. S. Siegler, W. Damon & R. M. Lerner (Eds.), *Handbook of child psychology: Cognition, perception, and language* (pp. 464–510). John Wiley & Sons.

- Sugisaki, K., & Isobe, M. (2001). Quantification without qualification without plausible dissent. *Semantics of Under-Represented Languages in the Americas*, 1(1), Article 13. <https://scholarworks.umass.edu/sula/vol1/iss1/13>
- Trueswell, J. C., Sekerina, I., Hill, N. M., & Logrip, M. L. (1999). The kindergarten-path effect: Studying on-line sentence processing in young children. *Cognition*, 73(2), 89–134.
- Trueswell, J. C., & Tanenhaus, M. K. (Eds.). (2005). *Processing world-situated language: Bridging the language-as-action and language-as-product traditions*. MIT Press.
- Woodard, K., Pozzan, L., & Trueswell, J. C. (2016). Taking your own path: Individual differences in executive function and language processing skills in child learners. *Journal of Experimental Child Psychology*, 141, 187–209. <https://doi.org/10.1016/j.jecp.2015.08.005>

APPENDIX

A. Full summary of LMM model of Total Fixation Duration

Fixed effect	Estimate	se	t	p
(Intercept)	0.47	0.08	5.62	<.001
Children	-0.04	0.08	-0.48	.629
Bunny	0.09	0.07	1.26	.208
Q-spreading	0.26	0.07	3.76	<.001
Extra	-0.17	0.07	-2.50	.013
TV	0.08	0.11	0.74	.463
IOI 2	-0.06	0.07	-0.85	.395
IOI 3	-0.03	0.07	-0.37	.711
IOI 4	-0.10	0.07	-1.47	.142
vocaS_c	0.01	0.00	1.94	.055
Children * Bunny	-0.08	0.11	-0.76	.446
Children * Q-spreading	0.03	0.11	0.23	.817
Children * Extra	0.11	0.11	1.02	.306
Bunny * Extra	0.01	0.10	0.13	.895
Q-spreading * Extra	-0.10	0.10	-1.05	.293
Children * TV	-0.11	0.11	-1.01	.312
Bunny * TV	-0.16	0.1	-1.60	.109
Q-spreading * TV	-0.20	0.1	-2.08	.037
Extra * TV	-0.03	0.1	-0.31	.760
Children * IOI 2	-0.02	0.11	-0.20	.844
Children * IOI 3	-0.05	0.11	-0.43	.667
Children * IOI 4	0.04	0.11	0.40	.689

Bunny * IOI 2	0.12	0.10	1.21	.226
Q-spreading * IOI 2	0.18	0.10	1.80	.072
Bunny * IOI 3	0.11	0.10	1.13	.257
Q-spreading * IOI 3	0.14	0.10	1.45	.148
Bunny * IOI 4	0.10	0.10	1.01	.314
Q-spreading * IOI 4	0.11	0.10	1.17	.242
Extra * IOI 2	0.54	0.10	5.46	<.001
Extra * IOI 3	0.35	0.10	3.53	<.001
Extra * IOI 4	0.53	0.10	5.43	<.001
TV * IOI 2	-0.06	0.10	-0.62	.535
TV * IOI 3	-0.09	0.10	-0.90	.367
TV * IOI 4	-0.05	0.10	-0.53	.597
Children * Bunny * Extra	-0.03	0.16	-0.21	.836
Children * Q-spreading * Extra	-0.26	0.16	-1.70	.090
Children * Bunny * TV	0.17	0.16	1.07	.283
Children * Q-spreading * TV	0.08	0.16	0.54	.593
Children * Extra * TV	0.03	0.16	0.22	.830
Bunny * Extra * TV	0.14	0.14	1.00	.315
Q-spreading * Extra * TV	-0.03	0.14	-0.23	.820
Children * Bunny * IOI 2	-0.12	0.16	-0.75	.453
Children * Q-spreading * IOI 2	-0.16	0.16	-1.02	.310
Children * Bunny * IOI 3	-0.07	0.16	-0.47	.639
Children * Q-spreading * IOI 3	-0.14	0.16	-0.89	.375
Children * Bunny * IOI 4	-0.14	0.16	-0.92	.359
Children * Q-spreading * IOI 4	-0.14	0.16	-0.89	.372
Children * Extra * IOI 2	-0.10	0.15	-0.67	.503
Children * Extra * IOI 3	0.07	0.15	0.45	.653
Children * Extra * IOI 4	-0.24	0.15	-1.56	.119
Bunny * Extra * IOI 2	-0.71	0.14	-5.09	<.001
Q-spreading * Extra * IOI 2	-0.73	0.14	-5.29	<.001
Bunny * Extra * IOI 3	-0.62	0.14	-4.50	<.001
Q-spreading * Extra * IOI 3	-0.57	0.14	-4.16	<.001
Bunny * Extra * IOI 4	-0.54	0.14	-3.93	<.001
Q-spreading * Extra * IOI 4	-0.58	0.14	-4.19	<.001
Children * TV * IOI 2	0.12	0.16	0.80	.422
Children * TV * IOI 3	0.16	0.16	1.01	.311
Children * TV * IOI 4	0.06	0.16	0.39	.695
Bunny * TV * IOI 2	0.11	0.14	0.77	.440
Q-spreading * TV * IOI 2	0.06	0.14	0.43	.670
Bunny * TV * IOI 3	0.07	0.14	0.51	.610
Q-spreading * TV * IOI 3	0.07	0.14	0.50	.617
Bunny * TV * IOI 4	0.07	0.14	0.48	.635
Q-spreading * TV * IOI 4	0.04	0.14	0.27	.787
Extra * TV * IOI 2	-0.08	0.14	-0.55	.583
Extra * TV * IOI 3	0.05	0.14	0.34	.731
Extra * TV * IOI 4	-0.06	0.14	-0.41	.684

Children * Bunny * Extra * TV	-0.01	0.22	-0.05	.963
Children * Q-spreading * Extra * TV	0.16	0.22	0.71	.480
Children * Bunny * Extra * IOI 2	0.51	0.22	2.34	.019
Children * Q-spreading * Extra * IOI 2	0.60	0.22	2.73	.006
Children * Bunny * Extra * IOI 3	0.29	0.22	1.32	.188
Children * Q-spreading * Extra * IOI 3	0.46	0.22	2.07	.038
Children * Bunny * Extra * IOI 4	0.49	0.22	2.23	.026
Children * Q-spreading * Extra * IOI 4	0.55	0.22	2.52	.012
Children * Bunny * TV * IOI 2	-0.04	0.22	-0.20	.842
Children * Q-spreading * TV * IOI 2	-0.16	0.22	-0.75	.456
Children * Bunny * TV * IOI 3	-0.07	0.22	-0.31	.759
Children * Q-spreading * TV * IOI 3	-0.13	0.22	-0.57	.566
Children * Bunny * TV * IOI 4	0.03	0.22	0.13	.894
Children * Q-spreading * TV * IOI 4	-0.11	0.22	-0.49	.628
Children * Extra * TV * IOI 2	-0.19	0.22	-0.87	.384
Children * Extra * TV * IOI 3	-0.35	0.22	-1.60	.109
Children * Extra * TV * IOI 4	-0.07	0.22	-0.31	.754
Bunny * Extra * TV * IOI 2	0.13	0.20	0.66	.512
Q-spreading * Extra * TV * IOI 2	0.09	0.20	0.44	.659
Bunny * Extra * TV * IOI 3	0.17	0.20	0.86	.391
Q-spreading * Extra * TV * IOI 3	0.05	0.20	0.27	.790
Bunny * Extra * TV * IOI 4	0.11	0.20	0.58	.560
Q-spreading * Extra * TV * IOI 4	0.11	0.20	0.54	.588
Children * Bunny * Extra * TV * IOI 2	-0.13	0.31	-0.41	.683
Children * Q-spreading * Extra * TV * IOI 2	0.29	0.31	0.92	.357
Children * Bunny * Extra * TV * IOI 3	0.05	0.31	0.14	.885
Children * Q-spreading * Extra * TV * IOI 3	0.21	0.31	0.67	.504
Children * Bunny * Extra * TV * IOI 4	-0.20	0.31	-0.65	.513
Children * Q-spreading * Extra * TV * IOI 4	0.18	0.31	0.58	.561

B. Full summary of LMM model of Fixation Count

Fixed effect	Estimate	se	t	p
(Intercept)	0.47	0.08	5.51	<.001
Children	-0.03	0.08	-0.39	.695
Bunny	0.10	0.07	1.46	.143
Q-spreading	0.30	0.07	4.43	<.001
Extra	-0.13	0.07	-1.88	.061
TV	0.08	0.11	0.69	.493
IOI 2	-0.04	0.07	-0.57	.569
IOI 3	-0.01	0.07	-0.11	.914
IOI 4	-0.10	0.07	-1.52	.130
vocaS_c	0.01	0.00	1.89	.061
Children * Bunny	-0.12	0.11	-1.14	.256
Children * Q-spreading	-0.01	0.11	-0.09	.931
Children * Extra	0.07	0.11	0.67	.500

Bunny * Extra	-0.04	0.09	-0.42	.677
Q-spreading * Extra	-0.18	0.09	-1.93	.054
Children * TV	-0.11	0.11	-1.00	.317
Bunny * TV	-0.14	0.10	-1.46	.143
Q-spreading * TV	-0.22	0.10	-2.33	.020
Extra * TV	-0.01	0.09	-0.14	.885
Children * IOI 2	-0.03	0.11	-0.27	.789
Children * IOI 3	-0.04	0.11	-0.37	.710
Children * IOI 4	0.04	0.11	0.38	.708
Bunny * IOI 2	0.09	0.10	0.93	.354
Q-spreading * IOI 2	0.14	0.10	1.49	.136
Bunny * IOI 3	0.08	0.10	0.79	.427
Q-spreading * IOI 3	0.10	0.09	1.00	.317
Bunny * IOI 4	0.08	0.09	0.82	.410
Q-spreading * IOI 4	0.07	0.09	0.76	.449
Extra * IOI 2	0.46	0.10	4.83	<.001
Extra * IOI 3	0.29	0.10	3.02	.003
Extra * IOI 4	0.51	0.09	5.33	<.001
TV * IOI 2	-0.05	0.10	-0.58	.564
TV * IOI 3	-0.07	0.10	-0.71	.480
TV * IOI 4	-0.03	0.10	-0.33	.739
Children * Bunny * Extra	0.06	0.15	0.42	.676
Children * Q-spreading * Extra	-0.15	0.15	-0.98	.326
Children * Bunny * TV	0.19	0.15	1.27	.203
Children * Q-spreading * TV	0.11	0.15	0.72	.472
Children * Extra * TV	0.04	0.15	0.25	.801
Bunny * Extra * TV	0.13	0.13	0.95	.344
Q-spreading * Extra * TV	-0.01	0.13	-0.04	.968
Children * Bunny * IOI 2	-0.09	0.15	-0.59	.553
Children * Q-spreading * IOI 2	-0.11	0.15	-0.75	.455
Children * Bunny * IOI 3	-0.04	0.15	-0.24	.814
Children * Q-spreading * IOI 3	-0.13	0.15	-0.89	.375
Children * Bunny * IOI 4	-0.07	0.15	-0.49	.628
Children * Q-spreading * IOI 4	-0.09	0.15	-0.61	.541
Children * Extra * IOI 2	-0.07	0.15	-0.49	.627
Children * Extra * IOI 3	0.03	0.15	0.22	.822
Children * Extra * IOI 4	-0.21	0.15	-1.40	.163
Bunny * Extra * IOI 2	-0.59	0.13	-4.38	<.001
Q-spreading * Extra * IOI 2	-0.62	0.14	-4.56	<.001
Bunny * Extra * IOI 3	-0.48	0.13	-3.57	<.001
Q-spreading * Extra * IOI 3	-0.45	0.13	-3.39	.001
Bunny * Extra * IOI 4	-0.46	0.13	-3.43	.001
Q-spreading * Extra * IOI 4	-0.48	0.13	-3.58	<.001
Children * TV * IOI 2	0.11	0.15	0.74	.457
Children * TV * IOI 3	0.12	0.15	0.78	.435
Children * TV * IOI 4	0.05	0.15	0.34	.735

Bunny * TV * IOI 2	0.10	0.13	0.74	.459
Q-spreading * TV * IOI 2	0.06	0.13	0.48	.633
Bunny * TV * IOI 3	0.05	0.13	0.40	.687
Q-spreading * TV * IOI 3	0.08	0.13	0.62	.536
Bunny * TV * IOI 4	0.04	0.13	0.31	.756
Q-spreading * TV * IOI 4	0.05	0.13	0.36	.722
Extra * TV * IOI 2	-0.08	0.13	-0.60	.548
Extra * TV * IOI 3	0.01	0.13	0.08	.939
Extra * TV * IOI 4	-0.12	0.13	-0.86	.391
Children * Bunny * Extra * TV	-0.05	0.21	-0.23	.821
Children * Q-spreading * Extra * TV	0.06	0.21	0.29	.772
Children * Bunny * Extra * IOI 2	0.42	0.21	1.99	.047
Children * Q-spreading * Extra * IOI 2	0.47	0.21	2.19	.029
Children * Bunny * Extra * IOI 3	0.22	0.21	1.04	.300
Children * Q-spreading * Extra * IOI 3	0.48	0.21	2.24	.025
Children * Bunny * Extra * IOI 4	0.30	0.21	1.40	.161
Children * Q-spreading * Extra * IOI 4	0.42	0.21	1.99	.047
Children * Bunny * TV * IOI 2	-0.04	0.21	-0.17	.864
Children * Q-spreading * TV * IOI 2	-0.16	0.21	-0.76	.446
Children * Bunny * TV * IOI 3	-0.04	0.21	-0.18	.857
Children * Q-spreading * TV * IOI 3	-0.09	0.21	-0.40	.688
Children * Bunny * TV * IOI 4	-0.02	0.21	-0.11	.915
Children * Q-spreading * TV * IOI 4	-0.13	0.21	-0.60	.549
Children * Extra * TV * IOI 2	-0.18	0.21	-0.84	.399
Children * Extra * TV * IOI 3	-0.27	0.21	-1.26	.206
Children * Extra * TV * IOI 4	-0.07	0.21	-0.31	.754
Bunny * Extra * TV * IOI 2	0.12	0.19	0.62	.535
Q-spreading * Extra * TV * IOI 2	0.04	0.19	0.23	.819
Bunny * Extra * TV * IOI 3	0.15	0.19	0.78	.436
Q-spreading * Extra * TV * IOI 3	0.00	0.19	0.02	.981
Bunny * Extra * TV * IOI 4	0.15	0.19	0.81	.415
Q-spreading * Extra * TV * IOI 4	0.09	0.19	0.47	.639
Children * Bunny * Extra * TV * IOI 2	-0.14	0.30	-0.47	.639
Children * Q-spreading * Extra * TV * IOI 2	0.33	0.30	1.10	.274
Children * Bunny * Extra * TV * IOI 3	-0.04	0.30	-0.14	.891
Children * Q-spreading * Extra * TV * IOI 3	0.12	0.30	0.40	.689
Children * Bunny * Extra * TV * IOI 4	-0.07	0.30	-0.23	.816
Children * Q-spreading * Extra * TV * IOI 4	0.25	0.30	0.82	.412

C. Full summary of LMM model of Total Fixation Duration in Children

Fixed effect	Estimate	se	t	p
(Intercept)	0.31	0.09	3.5	<.001
Error	0.11	0.05	2.29	.022
Bunny	0.04	0.04	1.04	.298
Q-spreading	0.27	0.05	5.94	<.001

Extra	0.18	0.04	4.5	<.001
Vocabulary	0.01	0.01	1.32	.194
Error * Bunny	-0.12	0.07	-1.68	.094
Error * Q-spreading	-0.18	0.07	-2.53	.011
Error * Extra	-0.17	0.07	-2.51	.012
Bunny * Extra	-0.08	0.06	-1.4	.161
Q-spreading * Extra	-0.39	0.06	-6.24	<.001
Error * Bunny * Extra	0.1	0.09	1.04	.299
Error * Q-spreading * Extra	0.31	0.09	3.31	<.001

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